Fiscal Year:	FY 2017	**Task Last Updated:**	FY 02/05/2018
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Project Title:	Clinical Trial Testing enterade® for the Alleviation of Gastrointestinal Toxicity in Radiomimetic Treated Patients		
Division Name:	Human Research		
Program/Discipline:	NSBRI		
Program/Discipline--Element/Subdiscipline:	NSBRI--Radiation Effects Team		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	02062	**Congressional District:**	8
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	NSBRI-RFA-SMARTCAP
Start Date:	06/01/2016	**End Date:**	05/31/2017
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NCC 9-58-RE04401		
Performance Goal No.:			
Performance Goal Text:			
	The primary objective of the study will be to compare the incidence of NCI-CTCAE 4.03, grade 3 or higher gastrointestinal toxicity between the medical food product, Enterade, with standard of care versus placebo with standard of care following Melphalan conditioning. As of July 14, 2017, a total of 74 patients have been screened for the study. 7 subjects were consented and waiting to be enrolled. A total of 63 subjects have been enrolled in the study. 38 patients have completed the study and 5 are currently active on study. Out of the completed subjects, 7 patients are not evaluatable for the study endpoints per the mITT principal. We have 14 subjects who have not completed the study due to various complications. The accrual rate with both active institutions is 12 patients per month. Based on actual recruitment rate, we anticipate that the last patient in will be on November 2017. Nevertheless, we anticipate increasing accrual rates as a result of other studies closing; this will increase the number of patients eligible to go on this trial.		

Task Description:	Results: The primary endpoint was diarrhea CTCAE v4.0 Grade 3 or higher. The rate of the primary endpoint is 24.4%, which is slightly lower than the degree of toxicity expected (30%). Since this is a double blind study, toxicity attribution is unrelated. Out of the 48 subjects who received the study product, 28 presented some degree to diarrhea. The majority of the patients on the study had CTCAE Grade 2 or 3 Diarrhea. There was no diarrhea CTCAE v4.0 Grade 4 or 5 in the study population. This endpoint will be looked at in conjunction with stool volume as documentation of diarrhea and incontinence in this population is done by recording both number of actual bowel movements per day and volume of stool per bowel movement. On interim analysis, 1/3 of the patients who completed the study had diarrhea on Day +7 and Day +14. Incontinence in the population was 7.3% at Day +7, while 2.7% on Day +14. The only Grade 4 or 5 Toxicities were related to expected bone marrow suppression or transplant related complications (veno-occlusive disease). Compliance: When considering only the 38 patients who completed the treatment, the median days of supplement taken is 12 (the range goes from 1-21). When all patients enrolled are considered, the median days of supplement taken is 9. The dose of treatment is 2, 8 oz bottles a day, which corresponds to 480 ml/day. The percent of patients who drank 480 mls/day for 14 is 31%. The reasons cited for the patient not taking the study supplement included patient refusal due to nausea, poor apetite, mucositis, taste, and not wanting it. Massachussetts General Hospital appears to be having better compliance than Dana Farber, with a compliance of approximately 60%. Safety: No adverse events related to the investigational drug have been reported so far. One death was attributed to transplant related complications, and not related to the study product. Impact on findings on Primary Objective and aims: After the review of this interim analysis, we have encountered two main issues in regards to the degree of toxicity seen in the study population, and the low compliance that both institutions are currently experiencing. -Low Toxicity: The degree of GI toxicity that is seen on this report is drastically lower than the one expected per protocol. -Low Compliance: Compliance of 31% is somehow concerning. The first aspect we have noticed is that patients appear to be refusing the study product due to the fact that they are not tolerating thin fluids. Other are having mucositis, which interferes with their oral intake. GI toxicity needs to be evaluated against compliance, to make sure that the reason why patients are refusing the product is due to the fact that they are very ill. This interim report is showing a low GI toxicity among the subjects, which does NOT correlate with such a low compliance. Plan: The investigators have decided to include CTCAE v4.0 Grade 2 and higher as primary endpoint, in order to have an increase in primary endpoint %. Also, the investigators will evaluate the patient's daily oral intake diaries, and will correlate their acceptance to thin fluids with the intake of study product. Investigators will adjust the antiemetic regimen which should improve compliance.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The alkylating agent, Melphalan, remains the mainstay of conditioning for autologous HSCT in multiple myeloma and non-Hodgkin lymphoma patients. Gastrointestinal symptoms represent the most significant non-hematologic toxicity following high-dose Melphalan conditioning. These symptoms include mucositis, diarrhea, nausea, vomiting, and decreased appetite. High dose chemotherapy also induces protein catabolism, disruption of the gastrointestinal mucosa, and oxidative stress. Increased intestinal permeability facilitates translocation of microorganisms and endotoxins into the bloodstream, which can lead to neutropenic fevers and even sepsis. As such, Melphalan functions as a radiomimetic agent in its impact on the integrity of the gastrointestinal tract. The mucosal sequelae of Melphalan and the subsequent enteritis are similar to those induced by total body irradiation, with approximately 40% of patients experiencing CTCAE grade 2 or higher diarrhea following conditioning regimens containing Melphalan. Being able to mitigate the gastrointestinal toxicity created by melphalan will correlate with the toxic effects of space-radiation. This prospective, double-blinded, phase 2 randomized study is to assess the effectiveness of Enterade in reducing diarrhea associated with high dose Melphalan chemotherapy. The hypothesis is that Enterade when combined with standard supportive care will help maintain the small bowel's ability to retain fluids leading to a reduction in diarrhea volume, patient reported gastrointestinal symptoms, and improved quality of life after autologous HSCT with Melphalan conditioning when compared to standard supportive care alone.
Task Progress:	Study Overview/Timing/ Reasons for delay: The study was activated at Dana Farber Cancer Institute/Brigham and Women's Hospital on 10/5/16, and patient enrollment started in 10/19/16. The starting date was delayed 3 months due to Institutional Review Board (IRB) review and delay in approval. After 4 months of active patient enrollment at Dana Farber, the study was temporarily closed in 1/9/17 due to an unknown object that was found inside one of the bottles when the Institutional Pharmacy was rebottling the product to be dispensed to the patient. This institutional contamination caused a study closure of 3 months, while IRB approved a trial reopening. It was reopened on 4/20/2017, a total of 9 subjects who were active on study while the contamination had to be taken off from study and removed from sample size. Due to severe IRB delays, as well as the pharmacy contamination report, it was decided to bring a second institution to assist on patient accrual. After subcontracting and training, Massachusetts General Hospital started patient enrollment on 4/27/2017. Subject Accrual: As of July 14, 2017, a total of 74 patients have been screened for the study. 7 subjects were consented and waiting to be enrolled. A total of 63 subjects have been enrolled in the study. 38 patients have completed the study and 5 are currently active on study. Out of the completed subjects, 7 patients are not evaluatable for the study endpoints per the mITT principal. We have 14 subjects who have not completed the study due to various complications. The accrual rate with both active institutions is 12 patients per month. Based on actual recruitment rate, we anticipate that the last patient in will be on November 2017. Nevertheless, we anticipate increasing accrual rates as a result of other studies closing; this will increase the number of patients eligible to go on this trial.
Bibliography Type:	Description: (Last Updated:)